University of Geneva

Faculty of Science

Quantum Computing Lab 2

Selected Chapters

14x060

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Contents

Exercise 1

Prove that the binary entropy $H_{\text{bin}}(p)$ attains its maximum value of one at $p = 1/2$.

Suppose we have a binary variable X that follows a Bernoulli distribution $\mathcal{B}$ of parameter p.

$$\mathbb{P}(X = 0) = p$$
$$\mathbb{P}(X = 1) = 1 - p$$

The entropy is defined as:

$$\begin{aligned} H(X) &= -\textstyle\sum_{x\in\{0,1\}} \mathbb{P}(X = x)\log_2(\mathbb{P}(X = x)) \\ &= -\mathbb{P}(X = 0)\log_2(\mathbb{P}(X = 0)) - \mathbb{P}(X = 1)\log_2(\mathbb{P}(X = 1)) \\ &= -p\log_2(p) - (1-p)\log_2(1-p) \\ &= f(p) \end{aligned}$$

We want to find the parameter p that maximize the entropy.

To do that, we can compute the derivate of the f function according to the parameter p to find optima of the function. We have:

$$\begin{aligned} \frac{\partial f}{\partial p} &= \frac{\partial}{\partial p}(-p\log_2(p) - (1-p)\log_2(1-p)) \\ &= \frac{\partial}{\partial p}\left(-p\frac{\ln(p)}{\ln(2)} - (1-p)\frac{\ln(1-p)}{\ln(2)}\right) \\ &= -\frac{\ln(p)}{\ln(2)} + \frac{p}{p\ln(2)} + \frac{\ln(1-p)}{\ln(2)} - \frac{1-p}{(1-p)\ln(2)} \\ &= -\frac{\ln(p) + 1 + \ln(1-p) - 1}{\ln(2)} \\ &= -\frac{\ln(p) + \ln(1-p)}{\ln(2)} \\ &= \ln\left(\frac{1-p}{p}\right)\cdot\frac{1}{\ln(2)} \end{aligned}$$

The optimum is obtained for $\frac{\partial f}{\partial p} = 0$, so:

$$
\begin{aligned}
&\frac{\partial f}{\partial p} = 0 \\
&\iff \ln\left(\frac{1-p}{p}\right) \cdot \frac{1}{\ln(2)} = 0 \\
&\iff \ln\left(\frac{1-p}{p}\right) = 0 \\
&\iff \frac{1-p}{p} = e^0 \\
&\iff 1-p = p \\
&\iff p = \frac{1}{2}
\end{aligned}
$$

So the entropy of a binary variable attains its maximum value at $p = \frac{1}{2}$. The maximum value of entropy obtained for $p = \frac{1}{2}$ is:

$$
\begin{aligned}
f\left(\frac{1}{2}\right) &= -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \left(1 - \frac{1}{2}\right)\log_2\left(1 - \frac{1}{2}\right) \\
&= \frac{1}{2} + \frac{1}{2} \\
&= 1
\end{aligned}
$$

Exercise 2: Sub-additivity of the Shannon entropy

Show that

$$H(p(x,y) \parallel p(x)p(y)) = H(p(x)) + H(p(y)) - H(p(x,y)).$$

By definition of the Kullback-Leibler divergence, $H(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log\left(\frac{P(x)}{Q(x)}\right)$.

The entropy is defined as $H(p(x)) = -\sum_x p(x)\log(p(x))$. We know that log is stricly increasing, with undefined value for 0 and $\log(1) = 0$. So as probabilities are defined in $[0, 1]$, the log is always less or equal to zero. So the entropy is always greater or equal to zero.

So we have:

$$
\begin{aligned}
H(p(x,y) \parallel p(x)p(y)) &= \sum_x \sum_y p(x,y) \log\left(\frac{p(x,y)}{p(x)p(y)}\right) \\
&= \sum_x \sum_y p(x,y)[\log(p(x,y)) - \log(p(x)p(y))] \\
&= \sum_x \sum_y p(x,y) \log(p(x,y)) - \sum_x \sum_y p(x,y) \log(p(x)p(y)) \\
&= -H(p(x,y)) - \sum_x \sum_y p(x,y)[\log(p(x)) + \log(p(y))] \\
&= -H(p(x,y)) - \sum_x \sum_y p(x,y) \log(p(x)) - \sum_x \sum_y p(x,y) \log(p(y)) \\
&= -H(p(x,y)) - \sum_x p(x) \log(p(x)) - \sum_y p(y) \log(p(y)) \\
&= -H(p(x,y)) + H(p(x)) + H(p(y))
\end{aligned}
$$

Deduce that $H(X,Y) \leq H(X) + H(Y)$, with equality if and only if X and Y are independent random variables.

We have shown that $H(p(x,y) \parallel p(x)p(y)) = H(p(x)) + H(p(y)) - H(p(x,y))$.

As the entropy of discrete variables is always greater or equal to zero, we have:

$$
\begin{aligned}
& H(p(x,y) \parallel p(x)p(y)) \geq 0 \\
\iff & H(p(x)) + H(p(y)) - H(p(x,y)) \geq 0 \\
& \iff H(p(x)) + H(p(y)) \geq H(p(x,y))
\end{aligned}
$$

If $p(x)$ and $p(y)$ are independent, $p(x,y) = p(x)p(y)$.

So

$$
\begin{aligned}
H(p(x,y)) &= H(p(x)p(y)) \\
&= -\sum_x \sum_y p(x)p(y) \log(p(x)p(y)) \\
&= -\sum_x \sum_y p(x)p(y)[\log(p(x)) + \log(p(y))] \\
&= -\sum_x \sum_y p(x)p(y) \log(p(x)) - \sum_x \sum_y p(x)p(y) \log(p(y)) \\
&= -\sum_x p(x) \log(p(x)) - \sum_y p(y) \log(p(y)) \\
&= H(p(x)) + H(p(y))
\end{aligned}
$$

So $H(p(x)) + H(p(y)) \geq H(p(x,y))$ and if $p(x)$ and $p(y)$ are independent,

$$
H(p(x)) + H(p(y)) \geq H(p(x,y)) = H(p(x)) + H(p(y))
$$

Exercise 3: Generalised measurements can decrease entropy

Suppose a qubit is in the state ρ measured using the measurement operators $M_1 = |0\rangle\langle 0|$ and $M_2 = |0\rangle\langle 1|$. If the result of the measurement is unknown to us then the state of the system afterwards is $M_1 \rho M_1^\dagger + M_2 \rho M_2^\dagger$. Show that this procedure can *decrease* the entropy of the qubit.

We have the two following measurement operators:

- $M_1 = |0\rangle\langle 0| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
- $M_2 = |0\rangle\langle 1| = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

The density ρ is defined as follow:

$$\rho = \sum_i p_i \ |\psi_i\rangle\langle\psi_i|$$

with p_i the probability to be in the state $|\psi_i\rangle$. As p_i is a probability, $\sum_i p_i = 0$.

In our case, as we work with only one qubit, this qubit can be in state $|0\rangle$ or $|1\rangle$.

So we have:

$$\begin{aligned}
\rho &= \sum_i p_i \ |\psi_i\rangle\langle\psi_i| \\
&= p_0 \ |0\rangle\langle 0| + p_1 \ |1\rangle\langle 1| \\
&= p_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ \ 0) + p_1 \begin{pmatrix} 0 \\ 1 \end{pmatrix} (0 \ \ 1) \\
&= \begin{pmatrix} p_0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & p_1 \end{pmatrix} \\
&= \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix}
\end{aligned}$$

Suppose that the result of the measurement is unknown to us.

So the state of the system afterwards is $\rho' = M_1 \rho M_1^\dagger + M_2 \rho M_2^\dagger$.

So we have:

$$
\begin{aligned}
\rho' &= M_1 \rho M_1^\dagger + M_2 \rho M_2^\dagger \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_0 & 0 \\ 0 & p_1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} p_0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & p_1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} p_0 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} p_1 & 0 \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\
&= |0\rangle\langle 0|
\end{aligned}
$$

So as $\rho = p_0 \; |0\rangle\langle 0| + p_1 \; |1\rangle\langle 1|$ and $\rho' = |0\rangle\langle 0|$, it means that the quantity of information has decreased. Indeed, ρ' can only be in one state $|0\rangle$ in a deterministic way, meaning that the entropy of ρ' is zero whereas the entropy of ρ is greater or equal to zero. Indeed, we have:

- $H(\rho) = -\sum_i p_i \log_2(p_i) = -p_0 \log_2(p_0) - p_1 \log_2(p_1) \geq 0$ since $\log_2$ of variable less than 1 is negative and as p_i is a probability, p_i is always positive.
- $H(\rho') = -\sum_i p_i \log_2(p_i) = -1 \log_2(1) = 0$ since $\log_2(1) = 0$.

So this procedure can decrease the entropy of the qubit.

Exercise 4

Suppose $|\psi\rangle$ and $|\varphi\rangle$ are two orthogonal quantum states of a single qubit. Design a quantum circuit with two input qubits (the "data" and the "target" qubits), with the data qubit in either the state $|\psi\rangle$ or $|\varphi\rangle$, and the target qubit prepared in the standard state $|0\rangle$, which produces as output $|\psi\rangle \; |\psi\rangle$ or $|\varphi\rangle \; |\varphi\rangle$, depending on whether $|\psi\rangle$ or $|\varphi\rangle$ was input to the data qubit.

We build a quantum circuit such as:

- We have a data state that can take values $|\varphi\rangle$ or $|\psi\rangle$.
- We prepare a target state that takes the value $|0\rangle$.

So initialy, we have either the state $|\varphi 0\rangle$ or the state $|\psi 0\rangle$.

As $|\psi\rangle$ and $|\varphi\rangle$ are two orthogonal quantum states of a single qubit, they form an orthogonal basis $\{|\psi\rangle, \; |\varphi\rangle\}$.

So it means that we can find an operator U that remaps the states to the basis $\{|0\rangle, \; |1\rangle\}$. As U is an operator, U must respect the constraint $U^\dagger U = I$.

So we will have for instance: $U \otimes |\psi\rangle = |0\rangle$ and $U \otimes |\varphi\rangle = |1\rangle$.

After applying the operator U, the circuit will be in state either $|00\rangle$ or $|10\rangle$. Now, if we apply the operator CNOT, the state $|00\rangle$ remains the same whereas the state $|10\rangle$ will become $|11\rangle$ due to the construction of the operator. Indeed, the CNOT operator invert the state of the target qubit (the second) if the control qubit (the first) is equal to 1, else leave the target qubit unchanged.

So now, the circuit will be in state either $|00\rangle$ or $|11\rangle$.

Finally, we apply the inverse operator $U^\dagger$ to the two qubits to come back to the basis $\{|\psi\rangle\ |\varphi\rangle\}$.

The circuit can be drawn like in Figure 1.

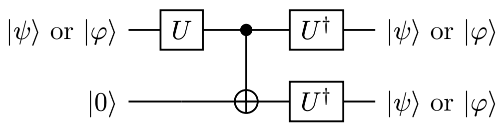

Figure 1: Quantum circuit to clone states of two orthogonal qubits

Exercise 5

Suppose Alice sends Bob an equal mixture of the four pure states

$$\begin{aligned}
|X_1\rangle &= |0\rangle \\
|X_2\rangle &= \sqrt{\frac{1}{3}}\left[|0\rangle + \sqrt{2}|1\rangle\right] \\
|X_3\rangle &= \sqrt{\frac{1}{3}}\left[|0\rangle + \sqrt{2}e^{2\pi i/3}|1\rangle\right] \\
|X_4\rangle &= \sqrt{\frac{1}{3}}\left[|0\rangle + \sqrt{2}e^{4\pi i/3}|1\rangle\right]
\end{aligned} \tag{1}$$

Show that the maximum mutual information between Bob's measurement and Alice's transmission is less than one bit. A POVM which achieves ≈ 0.415 bits is known. Can you construct this or better yet, one which achieves the Holevo bound?

Von Neumann entropy for quantum states

The entropy of a quantum state with density ρ is given by:

$$S(\rho) = -\operatorname{tr}(\rho \log \rho) \tag{2}$$

Theorem 11.8: Basic properties of von Neumann entropy

(1) The entropy is non-negative. The entropy is zero if and only if the state is pure.

(2) In a d-dimensional Hilbert space the entropy is at most $\log d$. The entropy is equal to $\log d$ if and only if the system is in the completely mixed state $\frac{I}{d}$.

(3) Suppose a composite system AB is in a pure state. Then $S(A) = S(B)$.

(4) Suppose p_i are probabilities, and the states ρ_i have support on orthogonal subspaces. Then

$$S\left(\sum_i p_i \rho_i\right) = H(p_i) + \sum_i p_i S(\rho_i). \tag{3}$$

(5) Joint entropy theorem: Suppose p_i are probabilities, $|i\rangle$ are orthogonal states for a system A, and ρ_i is any set of density operators for another system B. Then

$$S\left(\sum_i p_i \ |i\rangle\langle i| \otimes \rho_i\right) = H(p_i) + \sum_i p_i S(\rho_i). \tag{4}$$

Holevo bound theorem: Suppose Alice prepares a state ρ_x where $x = 0, ..., n$ with probabilities $p_0, ..., p_n$. Bob performs a measurement described by POVM elements $\{E_y\} = \{E_0, ..., E_m\}$ on that state, with measurement outcome Y.

The Holevo bound states that for any such measurement Bob may do:

$$H(X : Y) \leq S(\rho) - \sum_x p_x S(\rho_x) \tag{5}$$

where $\rho = \sum_x p_x \rho_x$.

As all the four pure states are only a composition of qubits $|0\rangle$ and $|1\rangle$, it means that they live in the $\{|0\rangle, |1\rangle\}$ basis which is an Hilbert basis.

In accordance with point (2) of von Neumann's properties of entropy, since the states lie in a two-dimensional Hilbert space, the entropy is at most $\log 2 = 1$ bit, with equality holding only if the system is in the fully mixed state $\frac{I}{2}$.

We want to find the bound of the mutual information according to the Holevo bound theorem.

Computation of $\sum_i p_i S(\rho_i)$

Alice send with probability $\frac{1}{4}$ one of the four pure states with density ρ_i. As she sends some pure state, according to property (1) of entropy of Von Neumann, the entropy is 0.

So we have:

$$\sum_i p_i S(\rho_i) = \sum_i p_i \cdot 0 = 0 \tag{6}$$

Precomputations

$$
\begin{aligned}
\rho_1 &= |X_1\rangle\langle X_1| \\
&= |0\rangle\langle 0| \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\
\rho_2 &= |X_2\rangle\langle X_2| \\
&= \sqrt{\frac{1}{3}}\Big[|0\rangle + \sqrt{2}|1\rangle\Big]\sqrt{\frac{1}{3}}\Big[\langle 0| + \sqrt{2}\langle 1|\Big] \\
&= \frac{1}{3}\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sqrt{2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]\Big[\begin{pmatrix} 1 & 0 \end{pmatrix} + \sqrt{2}\begin{pmatrix} 0 & 1 \end{pmatrix}\Big] \\
&= \frac{1}{3}\begin{pmatrix} 1 \\ \sqrt{2} \end{pmatrix}\begin{pmatrix} 1 & \sqrt{2} \end{pmatrix} \\
&= \frac{1}{3}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} \\
\rho_3 &= |X_3\rangle\langle X_3| \\
&= \sqrt{\frac{1}{3}}\Big[|0\rangle + \sqrt{2}e^{2\pi i/3}|1\rangle\Big]\sqrt{\frac{1}{3}}\Big[\langle 0| + \sqrt{2}e^{-2\pi i/3}\langle 1|\Big] \\
&= \frac{1}{3}\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sqrt{2}e^{2\pi i/3}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]\Big[\begin{pmatrix} 1 & 0 \end{pmatrix} + \sqrt{2}e^{-2\pi i/3}\begin{pmatrix} 0 & 1 \end{pmatrix}\Big] \\
&= \frac{1}{3}\begin{pmatrix} 1 \\ \sqrt{2}e^{2\pi i/3} \end{pmatrix}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \end{pmatrix} \\
&= \frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2e^{0} \end{pmatrix} \\
&= \frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix} \\
\rho_4 &= |X_4\rangle\langle X_4| \\
&= \sqrt{\frac{1}{3}}\Big[|0\rangle + \sqrt{2}e^{4\pi i/3}|1\rangle\Big]\sqrt{\frac{1}{3}}\Big[\langle 0| + \sqrt{2}e^{-4\pi i/3}\langle 1|\Big] \\
&= \frac{1}{3}\left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \sqrt{2}e^{4\pi i/3}\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]\Big[\begin{pmatrix} 1 & 0 \end{pmatrix} + \sqrt{2}e^{-4\pi i/3}\begin{pmatrix} 0 & 1 \end{pmatrix}\Big] \\
&= \frac{1}{3}\begin{pmatrix} 1 \\ \sqrt{2}e^{4\pi i/3} \end{pmatrix}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \end{pmatrix} \\
&= \frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}
\end{aligned} \tag{7}
$$

Computation of $S(\rho)$

If Alice sends Bob an equal mixture of the four pure states, it means that Bob will get one of the $|X_i\rangle$ with probability $p_i = \frac{1}{4}$.

So Bob can compute the density ρ of the system.

$$\begin{aligned}
\rho &= \sum_i p_i \rho_i \\
&= \frac{1}{4}\rho_1 + \frac{1}{4}\rho_2 + \frac{1}{4}\rho_3 + \frac{1}{4}\rho_4 \\
&= \frac{1}{4}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{12}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} + \frac{1}{12}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix} + \frac{1}{12}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix} \\
&= \frac{1}{12}\left(\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix} + \begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix} + \begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}\right) \\
&= \frac{1}{12}\begin{pmatrix} 6 & \sqrt{2}(1 + e^{-2\pi i/3} + e^{-4\pi i/3}) \\ \sqrt{2}(1 + e^{2\pi i/3} + e^{4\pi i/3}) & 6 \end{pmatrix} \\
&= \frac{1}{12}\begin{pmatrix} 6 & \sqrt{2}(e^{-6\pi i/3} + e^{-2\pi i/3} + e^{-4\pi i/3}) \\ \sqrt{2}(e^{6\pi i/3} + e^{2\pi i/3} + e^{4\pi i/3}) & 6 \end{pmatrix} \\
&= \frac{1}{12}\begin{pmatrix} 6 & \sqrt{2}(e^{(-2\pi i/3)^3} + e^{-2\pi i/3} + e^{(-4\pi i/3)^2}) \\ \sqrt{2}(e^{(6\pi i/3)^3} + e^{2\pi i/3} + e^{(4\pi i/3)^2}) & 6 \end{pmatrix} \\
&= \frac{1}{12}\begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}^{1} \\
&= \frac{I}{2}
\end{aligned} \tag{8}$$

So the entropy of Bob's system is maximal: $S(\rho) = S\left(\frac{I}{2}\right) = \log 2 = 1$.

So according to the Holevo bound theorem, we have:

$$\begin{aligned}
& H(X:Y) \leq S(\rho) - \sum_x p_x S(\rho_x) \\
\Longleftrightarrow\ & H(X:Y) \leq S(I/2) - 0 \\
\Longleftrightarrow\ & H(X:Y) \leq 1
\end{aligned} \tag{9}$$

Now we want to prove that the mutual information is stricly less than 1.

Mutual information is maximized only if Bob can determine Alice's qubit state without any error. Thus, if there is any overlap between the qubits $|X_i\rangle$, Bob might make a mistake, and the mutual information would not be maximized. Indeed, mutual information tells us to what extent knowing the state of Bob's system allows us to determine the state of Alice's system, and vice versa.

As the entropy of Bob's system is of 1, we want that the system of Bob allows to fully determinate the Alice's system to have a mutual information of 1.

So we want to verify whether the four qubits state do not overlap. However, we have basicaly only two dimensions, so we can only have two orthogonal states that are clearly distinguishable. So the four states must overlap, so Bob might make a mistake. So the mutual information is strictly less than 1.

[1] As $e^{2\pi i/3}$ and $e^{-2\pi i/3}$ are primitive root. Indeed, $e^{(2\pi i/3)^3} = e^{(-2\pi i/3)^3} = 1$ so $e^{2\pi i/3} + e^{(2\pi i/3)^2} + e^{(2\pi i/3)^3} = e^{-2\pi i/3} + e^{(-2\pi i/3)^2} + e^{(-2\pi i/3)^3} = 0$

The POVM (Posivite Operator-Valued Measure) is used for the analysis of the measurement. Indeed, suppose that a measurement is achieved with a measurement operator M_m on a quantum system in the state $|\psi\rangle$, then the probability of outcome m is given by

$$p(m) = \langle\psi|E_m|\psi\rangle \tag{10}$$

with $E_m = M_m^\dagger M_m$.

Due to the property of completeness equation of the measurement operator, we have:

$$\sum_m M_m^\dagger M_m = \sum_m E_m = I \tag{11}$$

On top of that, as $p(m)$ is a probability, $p(m) \geq 0\ \forall m$. So $\langle\psi|E_m|\psi\rangle \geq 0\ \forall E_m$, which means that E_m is a positif operator.

The complete set of operators $\{E_m\}$ is known as POVM.

So we want to construct a set of operators $\{E_m\}$ such that:

- Each operator E_m is positive
- $\sum_m E_m = I$

We want to check now if we can find a set of operators $\{E_m\}$ that are built from an orthogonal basis and that satisfy the two previous conditions. Indeed, the orthogonal basis maximize the recognition of quantum states while minimizing overlaps. Two states of an orthogonal basis don't overlap at all and are perfectly distinguishable, so they maximize the mutual information by minimizing the error of Bob's measurement.

We have the quantum pure states as described in Equation 1.

We can choose the basis $\{|0\rangle, |1\rangle\}$ which is a simple orthogonal basis.

Then, we can construct the POVM from this basis like follows:

$$\begin{aligned} E_1 &= |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \\ E_2 &= |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned} \tag{12}$$

Basically, we have $E_1 + E_2 = I$.

As used in page 534 of Quantum Computation and Quantum Information: 10th Anniversary Edition [1], the joint distribution $p(x, y)$ satisfies $p(x, y) = p(x)p(y|x) = p(x)\,\mathrm{tr}(\rho_x E_y)$.

So we can compute the table of the marginal and joint probabilities.

$x \setminus y$	1	2	$p(x)$
1	$p_1 \operatorname{tr}(\rho_1 E_1)$	$p_1 \operatorname{tr}(\rho_1 E_2)$	?
2	$p_2 \operatorname{tr}(\rho_2 E_1)$	$p_2 \operatorname{tr}(\rho_2 E_2)$	?
3	$p_3 \operatorname{tr}(\rho_3 E_1)$	$p_3 \operatorname{tr}(\rho_3 E_2)$	?
4	$p_4 \operatorname{tr}(\rho_4 E_1)$	$p_4 \operatorname{tr}(\rho_4 E_2)$	?
$p(y)$	?	?	1

=

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{4}\operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right)$	?
2	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right)$	?
3	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right)$	?
4	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{3}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right)$	?
$p(y)$	?	?	1

(13)

=

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{4}$	0	$\frac{1}{4}$
2	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$
3	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$
4	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{4}$
$p(y)$	$\frac{1}{2}$	$\frac{1}{2}$	1

So we have:

- $H(X) = -\sum_x p(x)\log_2(p(x)) = -4 \cdot \frac{1}{4}\log_2\left(\frac{1}{4}\right) = 2$
- $H(Y) = -\sum_y p(y)\log_2(p(y)) = -2 \cdot \frac{1}{2}\log_2\left(\frac{1}{2}\right) = 1$
- $H(X,Y) = -\sum_{x,y} p(x,y)\log_2(p(x,y)) = -\frac{1}{4}\log_2\left(\frac{1}{4}\right) - 3 \cdot \frac{1}{12}\log_2\left(\frac{1}{12}\right) - 3 \cdot \frac{1}{6}\log_2\left(\frac{1}{6}\right) = 2.689$

So the mutual information is:

$$H(X:Y) = H(X) + H(Y) - H(X,Y) = 2 + 1 - 2.689 = 0.311 \tag{14}$$

This is worse than the known POVM that achieves ≈ 0.415 bits.

So we can try another basis that is perhaps more suitable for this problem.

Indeed, the $\{|0\rangle, |1\rangle\}$ basis is not sensitive to phase whereas the basis $|+\rangle, |-\rangle$ is.

So we can construct the POVM from this basis like follows:

$$\begin{aligned} E_1 &= |+\rangle\langle+| = \frac{1}{\sqrt{2}}[|0\rangle + |1\rangle]\frac{1}{\sqrt{2}}[\langle 0| + \langle 1|] = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ E_2 &= |-\rangle\langle-| = \frac{1}{\sqrt{2}}[|0\rangle - |1\rangle]\frac{1}{\sqrt{2}}[\langle 0| - \langle 1|] = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \end{aligned} \tag{15}$$

So we have $E_1 + E_2 = I$.

Then, we can compute the joint- distribution $p(x, y)$:

$x \setminus y$	1	2	$p(x)$
1	$p_1 \operatorname{tr}(\rho_1 E_1)$	$p_1 \operatorname{tr}(\rho_1 E_2)$	?
2	$p_2 \operatorname{tr}(\rho_2 E_1)$	$p_2 \operatorname{tr}(\rho_2 E_2)$	?
3	$p_3 \operatorname{tr}(\rho_3 E_1)$	$p_3 \operatorname{tr}(\rho_3 E_2)$	?
4	$p_4 \operatorname{tr}(\rho_4 E_1)$	$p_4 \operatorname{tr}(\rho_4 E_2)$	?
$p(y)$	?	?	1

$=$

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\right)$	?
2	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2} \\ \sqrt{2} & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\right)$	?
3	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2}e^{-2\pi i/3} \\ \sqrt{2}e^{2\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\right)$	?
4	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\right)$	$\frac{1}{4}\operatorname{tr}\left(\frac{1}{6}\begin{pmatrix} 1 & \sqrt{2}e^{-4\pi i/3} \\ \sqrt{2}e^{4\pi i/3} & 2 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\right)$	?
$p(y)$	?	?	1

(16)

$=$

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{8}\operatorname{tr}\left(\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\right)$	$\frac{1}{8}\operatorname{tr}\left(\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}\right)$	?
2	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1+\sqrt{2} & 1+\sqrt{2} \\ 2+\sqrt{2} & 2+\sqrt{2} \end{pmatrix}\right)$	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1-\sqrt{2} & \sqrt{2}-1 \\ \sqrt{2}-2 & 2-\sqrt{2} \end{pmatrix}\right)$	?
3	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1+\sqrt{2}e^{-2\pi i/3} & 1+\sqrt{2}e^{-2\pi i/3} \\ 2+\sqrt{2}e^{2\pi i/3} & 2+\sqrt{2}e^{2\pi i/3} \end{pmatrix}\right)$	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1-\sqrt{2}e^{-2\pi i/3} & \sqrt{2}e^{-2\pi i/3}-1 \\ \sqrt{2}e^{2\pi i/3}-2 & 2-\sqrt{2}e^{2\pi i/3} \end{pmatrix}\right)$	?
4	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1+\sqrt{2}e^{-4\pi i/3} & 1+\sqrt{2}e^{-4\pi i/3} \\ 2+\sqrt{2}e^{4\pi i/3} & 2+\sqrt{2}e^{4\pi i/3} \end{pmatrix}\right)$	$\frac{1}{24}\operatorname{tr}\left(\begin{pmatrix} 1-\sqrt{2}e^{-4\pi i/3} & \sqrt{2}e^{-4\pi i/3}-1 \\ \sqrt{2}e^{4\pi i/3}-2 & 2-\sqrt{2}e^{4\pi i/3} \end{pmatrix}\right)$	?
$p(y)$	?	?	1

$=$

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
2	$\frac{1}{24}\left(3+2\sqrt{2}\right)$	$\frac{1}{24}\left(3-2\sqrt{2}\right)$	?
3	$\frac{1}{24}\left(3+2\sqrt{2}\right)$	$\frac{1}{24}\left(3-2\sqrt{2}\right)$	?
4	$\frac{1}{24}\left(3+2\sqrt{2}\right)$	$\frac{1}{24}\left(3-2\sqrt{2}\right)$	?
$p(y)$	?	?	1

$=$

$x \setminus y$	1	2	$p(x)$
1	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$
2	$\frac{1}{8}+\frac{\sqrt{2}}{12}$	$\frac{1}{8}-\frac{\sqrt{2}}{12}$	$\frac{1}{4}$
3	$\frac{1}{8}+\frac{\sqrt{2}}{12}$	$\frac{1}{8}-\frac{\sqrt{2}}{12}$	$\frac{1}{4}$
4	$\frac{1}{8}+\frac{\sqrt{2}}{12}$	$\frac{1}{8}-\frac{\sqrt{2}}{12}$	$\frac{1}{4}$
$p(y)$	$\frac{1}{2}+\frac{\sqrt{2}}{6}$	$\frac{1}{2}-\frac{\sqrt{2}}{6}$	1

So we have:

$$\begin{aligned}
H(X) &= -\textstyle\sum_x p(x)\log_2(p(x)) \\
&= -4\cdot\tfrac{1}{4}\log_2\left(\tfrac{1}{4}\right) \\
&= 2 \\
H(Y) &= -\textstyle\sum_y p(y)\log_2(p(y)) \\
&= -\left(\tfrac{1}{2}+\tfrac{\sqrt{2}}{6}\right)\log_2\left(\tfrac{1}{2}+\tfrac{\sqrt{2}}{6}\right)-\left(\tfrac{1}{2}-\tfrac{\sqrt{2}}{6}\right)\log_2\left(\tfrac{1}{2}-\tfrac{\sqrt{2}}{6}\right) \\
&= 0.833
\end{aligned}$$

$$\begin{aligned}
H(X,Y) &= -\textstyle\sum_{x,y} p(x,y)\log_2(p(x,y)) \\
&= -2\cdot\tfrac{1}{8}\log_2\left(\tfrac{1}{8}\right) - 3\cdot\left(\tfrac{1}{8}+\tfrac{\sqrt{2}}{12}\right)\log_2\left(\tfrac{1}{8}+\tfrac{\sqrt{2}}{12}\right) - 3\cdot\left(\tfrac{1}{8}-\tfrac{\sqrt{2}}{12}\right)\log_2\left(\tfrac{1}{8}-\tfrac{\sqrt{2}}{12}\right) \\
&= 2.390
\end{aligned}$$

So the mutual information is:

$$\begin{aligned}
H(X:Y) &= H(X)+H(Y)-H(X,Y) \\
&= 2+0.833-2.390 \\
&= 0.443
\end{aligned} \tag{17}$$

So we can see that the mutual information don't reach 1 bit, but the constructed POVM gives better results than the known POVM that gives a mutual information of 0.415 bits.

To archive all results (solutions, report together with your code, etc.) and submit them using Quantum computing course moodle.

Bibliography

[1] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010.